GRADE 3 · INTRO PYTHON TURTLE

Turning in Code

Ontario Math C3.1 — repeating events (move + turn)

Name: .... Date: ....

★ **I can read a loop that makes Bit move and turn.**

Move, then turn

Last time Bit moved in a straight line. Now we add a turn inside the loop, so Bit moves AND turns each time around.

The turn amount is given to you (a square corner is right 90). Read the loop and see the path Bit draws.

The block you know

repeat 3

- **move 50**
- **turn right 90**

Move 50, turn right 90 — three times.

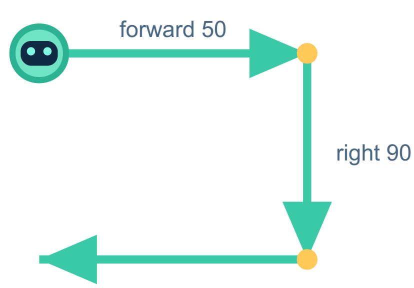
The same thing in Python

PYTHON

```
for step in range(3):
    bit.forward(50)
    bit.right(90)
```

OUTPUT

```
Bit draws 3 sides with 3 right-angle turns.
```



range(3) draws 3 sides and 3 turns — one more side would close a square.

1 Read it

In the loop, how many times does Bit move forward? ______ How many times does Bit turn? ______

2 Predict, then run

You change range(3) to range(4) (forward 50, right 90 stay the same). PREDICT what shape Bit draws now. Then run it to check.

3 Write your own

Make Bit move and turn right 90, FOUR times. Fill in the numbers: for step in range(______): bit.forward(50) bit.right(______)

4 Challenge

In for step in range(4): bit.forward(50); bit.right(90) — Bit turns 90 each time. How many degrees does Bit turn in TOTAL? (Hint: it goes all the way around.)

Big idea

Putting a move AND a turn inside the loop lets Bit draw shapes. The turn amount decides the corner; the number of repeats decides how many sides.